Process Control

Pre-Lab Plan Unit Operations Lab #2

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Group 2, Tuesday, PM section

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1. Experimental Objective (5 pts)

The objective of the experiment is to analyze the dynamic characteristics of a liquid level process and evaluate the effectiveness of feedback control mechanisms in controlling the tank level in the presence of disturbances and setpoint variations. The behavior of open loop is being characterized and compared with closed loop behavior using the PCT50 process control system to examine the effects of controller parameters on system stability, steady state offset, and transient behavior.

2. Experimental

The purpose of this experiment will be to investigate the dynamic behavior of a liquid level system and evaluate the performance of proportional and proportional-integral feedback control under disturbance and setpoint changes. The experiment will be conducted using the Armfield PCT50 level control apparatus. The system consists of a transparent cylindrical tank equipped with an inlet pump, adjustable control valves, a solenoid valve to introduce disturbances, and a pressure-based level sensor. The pump speed will serve as the manipulated variable, and the liquid level will serve as the primary process variable. The unit will be interfaced with a computer running the PCT50 software, which will allow real-time monitoring, controller tuning, and data acquisition. The software will record level and controller output continuously during each run for later analysis.

The experimental plan will be carried out in three phases. First, open loop testing will be performed to characterize baseline system dynamics. The pump speed will be manually adjusted to achieve steady state conditions, and disturbance tests will be introduced by opening and closing the solenoid valve. Fill and drain trials will also be conducted to observe transient behavior over a level range of approximately 0-100 mm. Next, closed loop proportional-integral control will be implemented with a 75 mm setpoint. The proportional band will be varied systematically to evaluate stability, steady state offset, and transient response. For each setting, disturbance responses and setpoint step tests will be conducted. The most stable and responsive proportional setting will be replicated to assess repeatability. Finally, proportional-integral control will be tested using the selected proportional setting. The integral time will varied stepwise, beginning at 100 s and decreasing until oscillatory or unstable behavior is observed. Disturbance and setpoint step responses will again be recorded to evaluate the effect of integral action on offset and stability.

Results will be analyzed by examining steady state offset, settling time, overshoot, and integrated error metrics. Replicated trials will be used to calculate mean values and 95% confidence intervals to quantify experimental variability. Linear fits and dynamic response modeling will be applied where appropriate to correlate experimental data with theoretical process control behavior. This analysis will allow comparison of controller performance and identification of optimal tuning parameters.

2.1 Apparatus (5 pts)

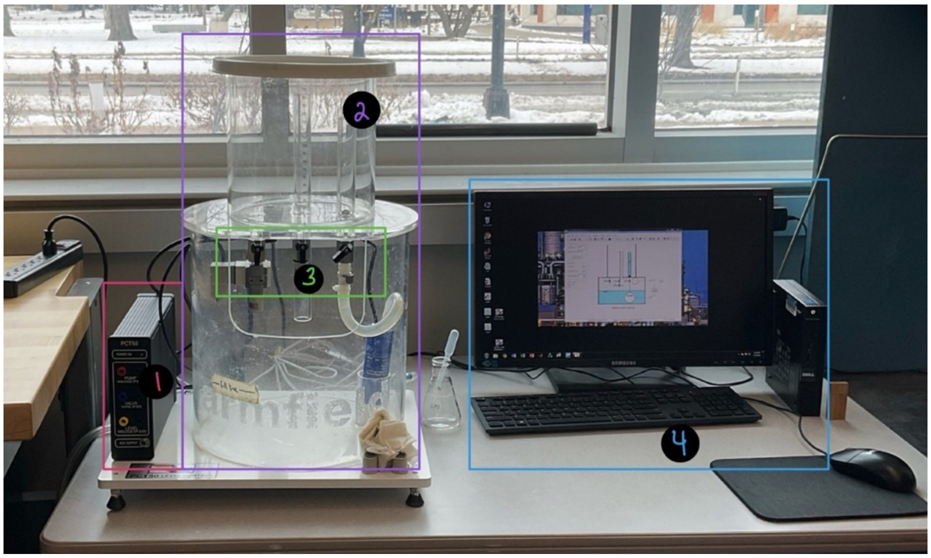

Figure 1: Apparatus with color coded labels, described in Table 1.

Table 1: Apparatus parts correlating to Figure 1

Number/ Color	Apparatus Part
1 – Pink	PCT50 PID Controller
2 – Purple	Water Reservoirs – with pump
3 – Green	Various valves and sensors
4 – Light Blue	Computer/Monitor with PCT50 Software

2.2 Materials and Supplies (5 pts)

Table 2: Materials and supplies used in experiment and their purpose

Materials/Supplies	Purpose
DI water	Water that will be used in reservoirs and sent between top and bottom via valves and pumps.
Beaker	Used to add DI water to reservoir initially
Drain tubing	Used to drain DI water from reservoir

2.3 Experimental Plan (10 pts)

- Fill the tank with DI water and ensure the pump is primed and the vessel can drain properly.
- Configure the level process for inflow control: CV2 fully open, install the 4 mm orifice under CV2 and 2 mm orifice under the solenoid valve.
- Start the PCT50 software, confirm PC communication, and open the mimic diagram for control/data collection.
- With the vessel empty, verify the level reads ~0 mm.
- Standardize the operating region: adjust CV1 so the level can hold near mid-height at ~50% pump speed, then return pump to 0% and drain back to empty before starting the formal runs.
- For experiment 1, start with an empty vessel.
- Increase pump speed gradually until the level is roughly steady. Log ~5 minutes at steady conditions.
- For disturbance test 1, open the solenoid valve, let level drop ≥ mm, then reduce pump speed to restore the original level for several minutes.
- Set pump speed to 0% and log the full drain to empty.
- For disturbance test 2, close the solenoid valve, let level rise ≥ 20 mm, then reduce pump speed to restore the original level.
- Set pump speed to 0% and log the full drain to empty.
- Fill/drain rate tests: fill from empty to 100 mm (at a higher pump speed than the steady test), stop the pump at 100 mm, and log the drain, repeat at 75% pump speed to 100 mm, stop, and drain.
- Replicate the 75% to 100 mm fill segment once for uncertainty on the fill-rate slope.
- For experiment 2, set controller mode to P-only (I = 0, D = 0) with setpoint 75 mm, start data acquisition.
- Run the following proportional band values in order: 200, 100, 50, 20, 10, 5, 2%.
- For each P setting:
    - Switch to automatic and let the response settle, record steady level and any oscillation.
    - Open the solenoid valve, allow it to reach a new steady behavior, close the solenoid and allow recovery.
    - Step the setpoint 75 → 100 mm, allow response, then return 100 → 75 mm, record transient response.
- Identify the best P-only setting and replicate the full disturbance + setpoint-step sequence twice at that setting.
- For experiment 3, choose the stable P setting from experiment 2, enable PI control, setpoint 75 mm, start data acquisition.
- Set I = 100 s, allow settling, record response.
- Open the solenoid valve, record deviation and recovery after closing the solenoid.
- Step the setpoint 75 → 100 → 75 mm, record transient response.
- Reduce integral time in steps (start at 50 s, then 30 s, 20 s, 15 s, 10 s, etc.) until sustained oscillation/instability is observed and log each condition.

- Select the best PI setting and replicate the disturbance + setpoint-step sequence twice.

The independent variables are the pump speed, the controller parameters (proportional band P swept from 200% down to 2% and integral time reduced stepwise starting at 100 s, then 50 s, and smaller values until instability), disturbance state, and setpoint level. The primary dependent variable is measured level. Levels will be measured mainly over 0-100 mm, since several sequences explicitly fill to 100 mm and drain to empty, and setpoint steps use 75 ↔ 100 mm. Performance will be quantified using steady-state offset, transient metrics, and error integrals such as IAE/ISE computed from $e(t) = L_{sp} - L(t)$. The experimental data will be compared against engineering expectations: open-loop segments will support a simple dynamic process model, and closed loop results will be correlated to known control behavior.

To estimate and reduce uncertainty, a quick sensor check/calibration will be performed by comparing software level to the vessel scale at multiple levels and fitting a linear calibration (slope/intercept with R^2 as goodness of fit), and use replicates on key conditions: at least 2 trials for the 75% fill-to-100 mm segment, and 2 full replicates at the best performing P-only setting and best PI setting to capture run-to-run variation. For scalar outputs, ±95% confidence intervals using a t-interval will be reported. For steady-state noise/stability, the standard deviation of level over a defined steady window will be computed. For model fits, goodness of fit will be reported with R^2 and RMSE/SSE to show how well the model captures the measured response.

Project Safety Evaluation (10 pts)

Since there is no chemical reaction, no hazardous chemical, or mechanical hazards, the risk posed by this experiment is low. The lab does have various electrical components, so it is essential to avoid water from getting to where it shouldn't be. Dr. Zdunek mentioned checking the pump output before turning it on, because at full capacity it can shoot the DI water far out of the system. If this does occur, turn off any electrical equipment to minimize damage. Another thing to note, mostly for the safety of the equipment, is to avoid running the pump dry. Make sure the liquid level is always sufficient to not cause damage.

	Yes/No	
Potential electrical hazards?	Y	If yes, identify conditions: Pump, Electric w/ water
Potential mechanical hazards?	N	If yes, identify conditions:
Potential pressure hazards?	N	If yes, identify conditions:
Potential temperature hazards?	N	If yes, identify conditions:
Hazardous/toxic chemicals involved?	N	If yes, hazards involved:
Gloves required?	N	If yes, specify type:
MSDS reviewed by all team members?	N	List MSDS sheets reviewed:
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	RT	
Pressure (psia)	Atm	
Safety Controls	*yes/no*	*If yes, Provide description*
	Y	System shut off

I have read relevant background material for the Unit Operations Laboratory entitled: "Process Control: Level Control" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

signature of team member	date
	2/4/26
	2/4/26
	2/4/2026
Kendall Johnson	2/4/26

Figure 2: Safety chart filled out.